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1 Day 1. Classic binomial and multinomial identities

Definition 1.1. Factorial of n , denoted by $n!$, is the number of ways to put n distinct objects in a queue.

Definition 1.2. Binomial coefficient $\binom{n}{k}$ is the number of ways to select k distinct objects from n distinct objects.

Definition 1.3. For $n = a + b + c$ the multinomial coefficient $\binom{n}{a,b,c}$ is the number of ways to split n distinct objects into different boxes, where the first box contains a objects, the second box contains b objects, the third box contains c objects.

One may create multinomial coefficient for any number of boxes :)

Exercise 1.1. Is $\binom{n}{k}$ equal to $\binom{n}{n-k}$?

Is $\binom{n}{k}$ equal to $\binom{n}{k, n-k}$?

Exercise 1.2. Write the product

$$\binom{100}{20} \cdot \binom{80}{50} \cdot \binom{30}{10}$$

as one multinomial coefficient.

Exercise 1.3. Write

$$\binom{n}{a, b, c, d}$$

using binomial coefficients.

Write

$$\binom{n}{a, b, c, d}$$

using factorials of n, a, b, c and d .

Idea 1. Solution strategies for combinatorial identities:

1. Double counting or story proof.
 2. Visual identity (rare gem!).
 3. Evaluation of a polynomial at a special point. Differentiation, integration.
 4. Combine old identities into new.
 5. Guess the answer and verify by induction.
 6. Dark art!
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Pascal's triangle and rule

Exercise 1.4. Recall the Pascal arithmetic triangle. Draw two versions: triangular and rectangular.

Exercise 1.5. Pascal's rule. Simplify

$$\begin{aligned} & \binom{n}{k} + \binom{n}{k+1}. \\ & \binom{n}{a-1, b, c} + \binom{n}{a, b-1, c} + \binom{n}{a, b, c-1}. \\ & \binom{n}{a-1, b, c, d} + \binom{n}{a, b-1, c, d} + \binom{n}{a, b, c-1, d} + \binom{n}{a, b, c, d-1}. \end{aligned}$$

Exercise 1.6. Simplify

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

Recall the Pascal's rule and simplify

$$\binom{2n}{0} + \binom{2n}{2} + \binom{2n}{4} + \cdots + \binom{2n}{2n}.$$

Simplify

$$\binom{2n-1}{0} + \binom{2n-1}{2} + \binom{2n-1}{4} + \cdots + \binom{2n-1}{2n-2}.$$

Exercise 1.7. Simplify

$$\begin{aligned} & \sum_{a+b+c=2026} \binom{2026}{a, b, c}. \\ & \sum_{a+b+c+d=2027} \binom{2027}{a, b, c, d}. \\ & \sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}. \\ & \sum_{a+b \leq n} \binom{n}{a} \cdot \binom{n-a}{b}. \end{aligned}$$

Here a , b and c are assumed non-negative integers.

Hockey-stick identity

Exercise 1.8. Hockey-stick identity, Christmas stocking identity, boomerang identity.

Simplify

$$\begin{aligned} & \sum_{n=a}^b \binom{n}{a}. \\ & 1 + 2 + 3 + \cdots + n. \\ & 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n \cdot (n+1). \\ & 1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \cdots + n \cdot (n+1) \cdot (n+2). \end{aligned}$$

Exercise 1.9. Obtain closed form expressions for:

$$1^2 + 2^2 + \cdots + n^2,$$

$$1^3 + 2^3 + \cdots + n^3,$$

$$1^4 + 2^4 + \cdots + n^4.$$

Exercise 1.10. Hockey-sticks plus idea from linear algebra!

Decompose $n^2 + 2n + 5$ using $\binom{n}{0}$, $\binom{n}{1}$, $\binom{n}{2}$ as the basis.

Find the sum

$$\sum_{k=2}^{2026} k^2 + 2k + 5.$$

Using similar trick find the sums

$$\sum_{k=1}^n 5k^2 + 4k, \quad \sum_{k=1}^n k^3 + 2k^2.$$

Exercise 1.11. Let's start from some number $\binom{n}{k}$ inside the Pascal's triangle.

In one step we can go either in north-east or north-west direction.

Find the sum of all numbers we can reach.

Binomial or Taylor or MacLaurin expansion

Exercise 1.12. Write the Taylor expansion (binomial expansion) at $x = 0$ for

$$(1+x)^n, \quad (x+y)^n.$$

Exercise 1.13. By plugging some special points to the $(x+y)^n$ or somewhere else find the sums

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

$$\sum_{k=0}^n k \binom{n}{k}, \quad \sum_{k=0}^n (k-1) \binom{n}{k}, \quad \sum_{k=0}^n (-1)^k \binom{n}{k}, \quad \sum_{k=0}^n (-1)^{k-1} k \binom{n}{k}$$

Hint: differentiation may help you.

How that can be?

Exercise 1.14. Be brave, really brave!

Take a deep breath and calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

Exercise 1.15. Simplify

$$\binom{-1/2}{3} + \binom{-1/2}{4}.$$

Draw the Pascal's triangle starting from $\binom{-1/2}{0}$.

Draw the Pascal's triangle starting from $\binom{1/3}{0}$.

Exercise 1.16. Write the Taylor expansion at $x = 0$ for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

Exercise 1.17. Simplify

$$\binom{10}{0} \binom{-0.5}{7} + \binom{10}{1} \binom{-0.5}{6} + \binom{10}{2} \binom{-0.5}{5} + \cdots + \binom{10}{7} \binom{-0.5}{0}.$$

Hint: for a honest proof consider the binomial expansion of $(1+x)^1$, $(1+x)^{-0.5}$, $(1+x)^{10-0.5}$.

Exercise 1.18. Homage to Newton.

Write the Taylor expansion at $x = 0$ for $(1-x^2)^{1/2}$.

Visually (or analytically if you wish) find the integral

$$\int_{-1}^1 (1-x^2)^{1/2}.$$

Calculate the sums

$$\binom{1/2}{0} - \frac{1}{3} \binom{1/2}{1} + \frac{1}{5} \binom{1/2}{2} + \cdots$$
$$\sum_{k=2}^{\infty} \frac{1}{2k+1} \frac{(2k-3)!!}{(2k)!!}$$

Planned entrance quiz :)

1. How many ways are to order 6 persons in a queue?
2. The box contains 50 balls numbered from 1 to 50. We draw 6 balls without replacement and write down the resulting sequence of numbers. How many different sequences are possible?
3. I have 5 yellow books and 7 red books. I want to place them on my bookshelf and I would like to place yellow books to the left of red books. How many book orders are possible?

Home assignment-1:

1. 1.7(3) in the old enumeration:

Simplify

$$\sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}.$$

2. 1.9(2) in the old enumeration:

Find explicit formula for the sum

$$1^3 + 2^3 + \dots + n^3.$$

Hint: represent the general term k^3 as the sum of binomial coefficients and then use the hockey stick or Christmas stocking or boomerang!

3. 1.10(2) in the old enumeration:

[*] Using hockey stick decomposition find the sums

$$\sum_{k=1}^n 5k^2 + 4k.$$

4. 1.19 in the old enumeration:

[*] Take a deep breath and calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

5. 1.21 in the old enumeration:

Write the Taylor expansion at $x = 0$ for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

6. Watch a Veritasium video «Newton changed the game»:

<https://youtu.be/gMlf1ELvRzc?si=VASoVKQj3Pk-wstA>.

You are more than welcome to solve other exercises that we skipped. Starred exercises will be on the quiz next day.

2 Day 2. Vandermonde's identity and beyond

Vandermonde's identity

Exercise 2.1. Simplify

$$\begin{aligned} & \binom{a}{0} \binom{b}{m} + \binom{a}{1} \binom{b}{m-1} + \binom{a}{2} \binom{b}{m-2} + \cdots + \binom{a}{m} \binom{b}{0}. \\ & \binom{a}{0} \binom{m}{0} + \binom{a}{1} \binom{m}{1} + \binom{a}{2} \binom{m}{2} + \cdots + \binom{a}{m} \binom{m}{m}. \\ & \binom{a}{0}^2 + \binom{a}{1}^2 + \binom{a}{2}^2 + \cdots + \binom{a}{a}^2. \end{aligned}$$

Exercise 2.2. There are $n = 500$ fishes in the pond, $t = 30$ of them are tagged. You capture $c = 5$ fishes at random without replacement. Let the random variable Y be the number of captured tagged fishes. Find the probabilities $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$, $\mathbb{P}(Y = y)$.

Exercise 2.3. There are $n = 24$ candies in the jar, 14 with orange flavour and 10 with ananas flavour. You take 6 candies at random. Let Y be the number of candies with orange flavour you picked. Find the probabilities $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$, $\mathbb{P}(Y = y)$.

Exercise 2.4. A bridge hand (random 13 cards out of 52-card standard deck) is dealt.

What is the probability that the hand contains no spades?

What is the probability that the hand contains exactly 5 hearts?

What is the probability that the hand contains at most two aces?

Exercise 2.5. Simplify

$$\sum_{a+b+c=k} \binom{n_1}{a} \binom{n_2}{b} \binom{n_3}{c}.$$

Gauss summation trick

Exercise 2.6. Simplify

$$\begin{aligned} & 1 + 2 + 3 + \cdots + n; \\ & 1^2 + 2^2 + 3^2 + \cdots + n^2; \\ & \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}; \\ & \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \cdots + (n+1)\binom{n}{n}; \\ & \binom{n}{2} + 2\binom{n}{3} + 3\binom{n}{4} + \cdots + (n-1)\binom{n}{n}; \\ & \binom{n}{0} + 3\binom{n}{1} + 5\binom{n}{2} + \cdots + (2n+1)\binom{n}{n}; \\ & \binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - 4\binom{n}{3} + \cdots + (-1)^n(n+1)\binom{n}{n}; \end{aligned}$$

$$3\binom{n}{1} + 7\binom{n}{2} + 11\binom{n}{3} + \cdots + (4n-1)\binom{n}{n};$$
$$\binom{n}{1} - 2\binom{n}{2} + 3\binom{n}{3} - 4\binom{n}{4} + \cdots + (-1)^n n\binom{n}{n};$$

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